Project Report: A Data-Driven Strategy for a New Movie Studio

This report provides an in-depth analysis based on the provided objectives and the findings from the notebooks (Talanta-hela). It outlines a data-driven strategy for a new movie studio, focusing on identifying key success factors, understanding high-performing films, and leveraging insights for decision-making.

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1. Business Understanding

The core objective of this project is to develop a data-driven content strategy for a new movie studio to help it succeed in a highly competitive market. The primary stakeholder is the movie studio's leadership, who need to make informed decisions on film selection, budgeting, and marketing. The project addresses the real-world problem of mitigating the financial risk inherent in film production by identifying key factors that lead to commercial success.

2. Data Understanding

The analysis leverages a comprehensive dataset created by merging five different data sources. The data is suitable for the project as it combines financial data, critical and audience reception, and key movie attributes.

- **Data Sources:** The project utilized data from Box Office Mojo (bom.movie_gross.csv), IMDb (im.db), Rotten Tomatoes (rt.movie_info.tsv), TheMovieDB (tmdb.movies.csv), and The Numbers (tn.movie_budgets.csv).

1. tn.movie_budgets

- **id**: Unique identifier for the movie entry.
- **release_date**: The official release date of the movie.
- **movie**: Title of the movie.
- **production_budget**: Total cost of producing the movie.
- **domestic_gross**: Total revenue generated in the domestic (U.S.) market.
  worldwide_gross: Total revenue generated globally.

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2. bom.movie_gross

- **title**: Name of the movie.
- **studio**: The studio or production company that released the movie.

- **domestic_gross**: Domestic box office revenue.
- **foreign_gross**: International box office revenue.
- **year**: Release year of the movie.

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3. rt.movie_info

- **id**: Unique identifier (likely corresponding to Rotten Tomatoes).
- **synopsis**: A short description or summary of the film's plot.
- **rating**: Content rating (e.g., PG-13, R).
- **genre**: Primary genre or genres of the movie.
- **director**: Name of the director.
- **writer**: Name of the screenwriter.
- **theater_date**: Date when the movie was released in theaters.
- **dvd_date**: Date when the movie was released on DVD.
- **currency**: Currency used in the box office value.
- **box_office**: Box office revenue.
- **runtime**: Duration of the movie.
- **studio**: Producing studio or distributor.

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4. rt.reviews

- **id**: Unique identifier linking to the movie being reviewed.
- **review**: Text content of the critic's review.
- **rating**: Numeric or qualitative score given by the critic.
- **fresh**: Indicator of whether the review was "fresh" (positive) or "rotten" (negative).
- **critic**: Name of the critic.
- **top_critic**: Boolean flag or indicator for top-tier critics.
- **publisher**: Publisher or outlet the review appeared in.
- **date**: Date the review was published.

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5. tmdb.movies

- **Unnamed: 0**: Row index (can be ignored).

- **genre_ids**: Numerical ID(s) representing the genre(s) of the movie.
- **id**: Unique identifier from TMDB.
- **original_language**: Original language of the film.
- **original_title**: The original title as released.
- **popularity**: Popularity score as computed by TMDB.
- **release_date**: Movie release date.
- **title**: Final displayed title.
- **vote_average**: Average rating from users.
- **vote_count**: Number of votes received.

6. IMDb SQLite Database (im.db)

- **Tables**: 8 (movie_basics, directors, known for, movie_akas, movie_ratings, persons, principals, writers)
- **Key Tables Analyzed**:
    - **movie_basics**: Contains movie_id, primary title, original_title, start year, runtime minutes, genres.
    - **movie_ratings**: Contains movie_id, average rating, numvotes.
- **Key Observations**:
    - movie_basics has 146,144 rows with missing values in runtime minutes and genres.
    - movie_ratings has 73,856 rows with ratings and vote counts, but not all movies in movie_basics have ratings.
    - movie_id can be used to link with other datasets (e.g., TMDb) for enriched analysis.
    - Useful for detailed movie metadata and ratings.

- **Properties and Relevance:** This combination of datasets allows for a holistic view of a film's performance, from financial metrics like worldwide_gross and production_budget to qualitative metrics such as averageRating and numVotes. The notebooks justify the use of these datasets for their ability to provide the necessary features to analyze movie success.
- **Limitations:** A key limitation is the potential for data loss during the merging process, as not all movies are present across all datasets.

3. Data Preparation

1. tn.movie_budgets (The Numbers Dataset)

This dataset required extensive cleaning due to the presence of currency formatting in financial columns.

- **Removed characters** such as $ and , from production_budget, domestic_gross, and worldwide_gross.
- **Converted** these columns from string to numeric (float) for proper mathematical operations.
- **Dropped rows** where essential financial data was missing (e.g., null or zero production_budget or worldwide_gross).
- **Ensured release_date** was converted to datetime format for time-based analysis.

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2. bom.movie_gross (Box Office Mojo Dataset)

This dataset was lightly cleaned:

- **Removed commas and dollar signs** from domestic_gross and foreign_gross columns.
- **Converted** these fields into numeric types.
- **Filtered rows** to ensure valid and complete entries for title, studio, and year.
- **Standardized title casing** to help with later joins across datasets.

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3. rt.movie_info (Rotten Tomatoes Movie Metadata)

This TSV dataset required formatting and type consistency improvements:

- **Cleaned missing or malformed values** in columns like box_office, genre, and rating.
- **Standardized genres** (e.g., converting lists or pipe-separated values to more consistent strings).
- **Dropped non-informative columns** such as synopsis if not used in the analysis.
- **Filtered duplicate or unnamed rows** that had no usable movie identifier or empty id.

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4. rt.reviews (Rotten Tomatoes Reviews)

This dataset provided user/critic review data and was cleaned as follows:

- **Removed null values** in review, critic, or rating columns.
- **Mapped values** in fresh to binary format (e.g., Fresh = 1, Rotten = 0) where used.
- **Joined** with movie info via the id field, so irrelevant entries (with missing id) were removed.

5. tmdb.movies (The Movie Database Dataset)

This dataset contained many technical fields, requiring:

- **Dropped redundant or low-utility columns** such as Unnamed: 0, which was not used in the analysis.
- **Exploded the genres column** (initially a stringified list of dictionaries or IDs) to analyze genres more granularly.

- **Standardized date formatting** in release_date and ensured proper datetime conversion.
- **Checked and cleaned null values** in important columns like title, popularity, and vote_average.

Merging and Final Cleaning

Once the individual datasets were cleaned:

- **Merged on common identifiers**, such as cleaned movie title and release_year, or id where available.
- **Feature engineered columns** like profit = worldwide_gross - production_budget and roi = profit / production_budget.
- **Removed duplicate rows** after merge.
- **Handled outliers** by identifying extremely low or negative profits (especially for ROI analysis).
- **Ensured type consistency** across merged columns for financial and rating fields.

Data Validation

Introduction

The primary objective of this phase was to assess the **quality, consistency, and integrity** of the raw dataset before proceeding with in-depth analysis and insights recommendation. A robust data validation process is crucial to ensure that subsequent insights are reliable and actionable.

Data Source and Overview

The datasets under examination were

1. **Tmdb_tn_merged** which had 5940 rows × 6 columns. The examined columns were id, title, production budget, domestic gross, worldwide gross and release year

2. **Rt_combined_df** dataset contained 1560 rows and 12 columns. The columns of interest in this dataset were: Id, Synopsis, Rating', Genre', Director', Writer', Runtime', Review counts, Freshness score, Theatre and DVD date.
3. **The top_writers_by_review** dataset was 10 rows and 4 columns. The columns in validated here were writer, writer_avg_freshness, writer_avg_reviews, writer_movie_count.
4. **The top_directors_by_review** dataset was 10 rows and had these 4 columns: director, director_avg_freshness, director_avg_reviews, director_movie_count

Validation Objectives

The data validation stage focused on identifying and addressing the following data quality issues:

1. **Completeness**: Checking for missing values (nulls) across all columns.
2. **Uniqueness**: Verifying the uniqueness of key identifiers.
3. **Consistency**: Ensuring data types are appropriate and values adhere to expected formats (e.g., dates, numerical ranges).
4. **Validity**: Confirming that values fall within acceptable ranges or predefined categories.

Methodology

The data validation process involved a combination of automated scripts and manual inspection:

1. **Initial Data Profiling**: Used descriptive statistics .info(), .describe(), .nunique() to get a high-level overview of data types, non-null counts, and basic statistical measures.
2. **Missing Value inspection**: identified missing values per column and investigated patterns using .isnull().sum().
3. **Duplicate Record Detection**: Checked for duplicate entries using .duplicated with the id column as the subset.
4. **Data Type Verification and Conversion**: Ensured date columns like theatre_date were datetime objects and numerical columns were appropriate types (e.g., float, int) using .dtypes
5. **Outlier Detection**: Employed IQR (Interquartile Range) method for all numerical columns and visual inspection (histograms, box plots) to identify extreme values.
6. **Data Range and Format Checks**: Ensured numerical data were within the was positive where applicable using density plots.

Key Findings

The data validation process revealed several issues requiring attention:

1. **Missing Data Handling**:

- Tmdb_tn_merged had no missing values
- Rt combined had missing values in the Review counts, Freshness Score, Theatre and DVD date. To Impute the null, the review counts were filled with 0 and the Freshness Score was imputed using the median. The Theatre date and DVD date columns were dropped because this analysis only focused on the release year.
- Top writers and Top directors had no missing values.

2. **Duplicate Resolution**:

- The movie id duplicates were kept as they represented different movie titles but contained the same movie id.

3. **Data Cleaning and Transformation**:

- For all numerical values in the datasets, the interquartile range was found and the outliers excluded from the analysis as they may skew the statistical description of the dataset.
- After validation all numerical values produced a standard distribution.

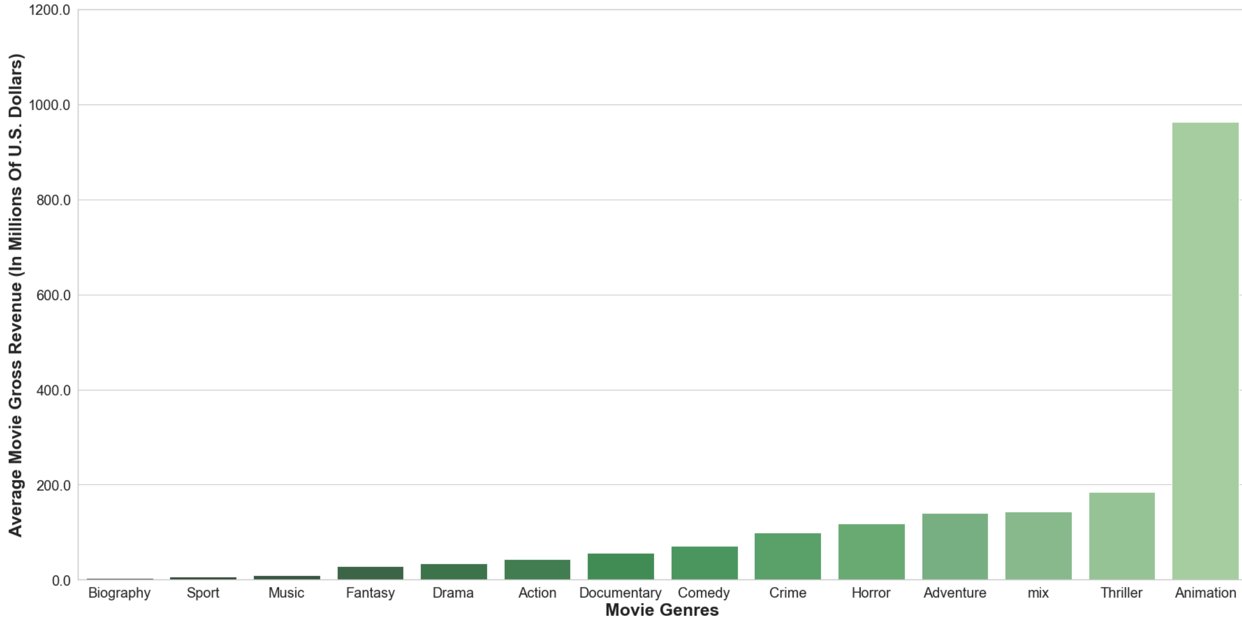
The data validation stage identified data quality issues within the datasets which were addressed through imputing ,dropping rows and columns..

4. Data Analysis & Findings

The analysis explored several key areas to understand the drivers of movie success.

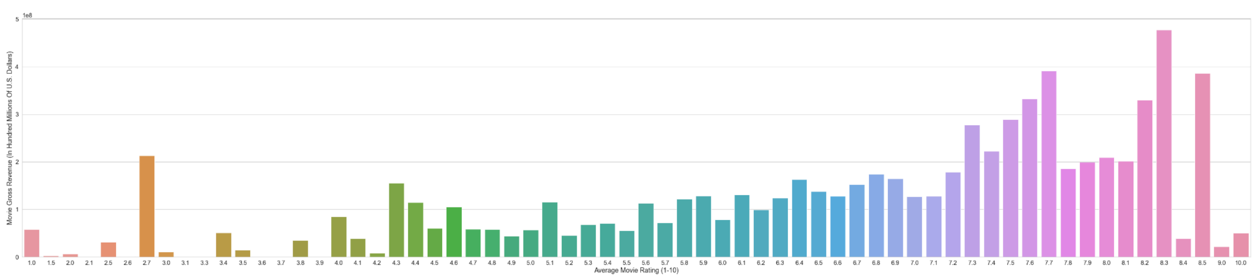
High-Performing Films & Genres

The analysis of high-performing films reveals that certain genres consistently achieve higher profits. The notebooks show that Thriller and mix films, as well as those with a combination of genres, tend to have the highest average profit.



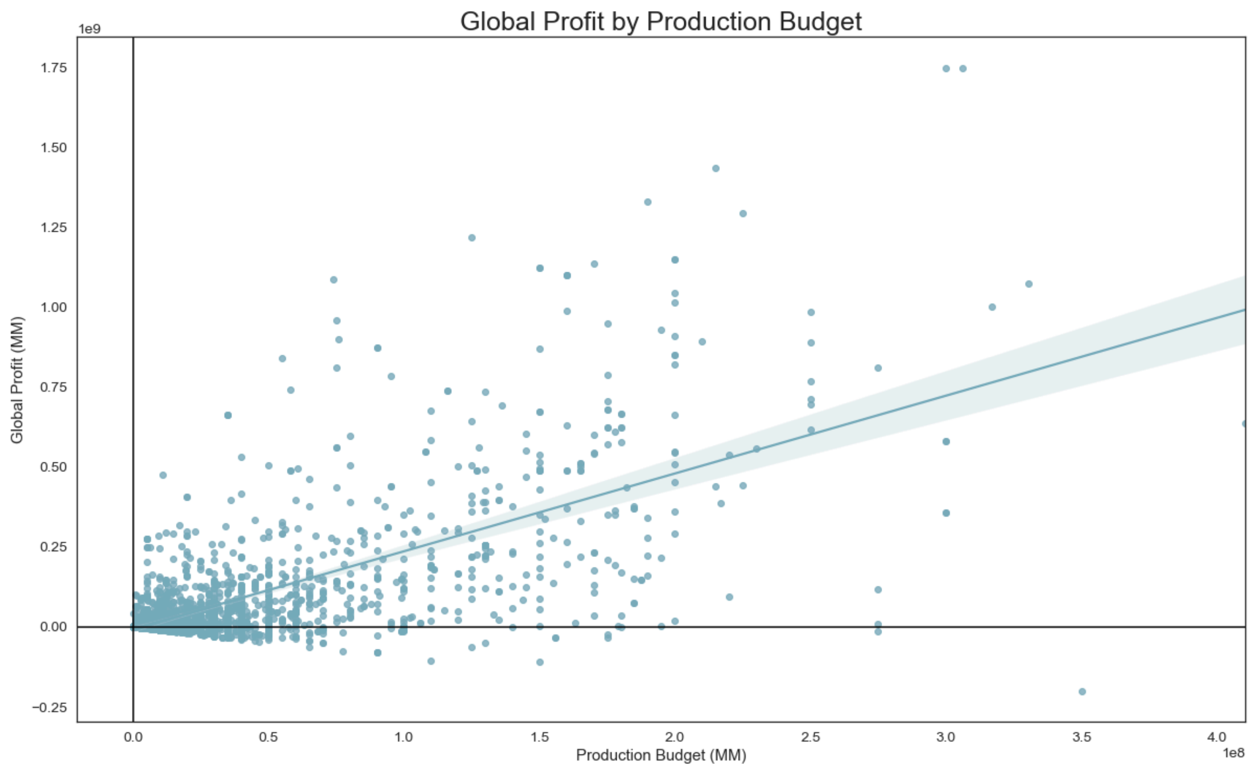
Relationship Between Critical Reception and Financial Success

There is a strong, positive correlation between a movie's ratings and its financial performance. The notebooks' analysis confirmed that films with higher IMDb ratings generally have a significantly higher worldwide gross. An ANOVA test concluded that the difference in average profit across various rating categories is statistically significant.



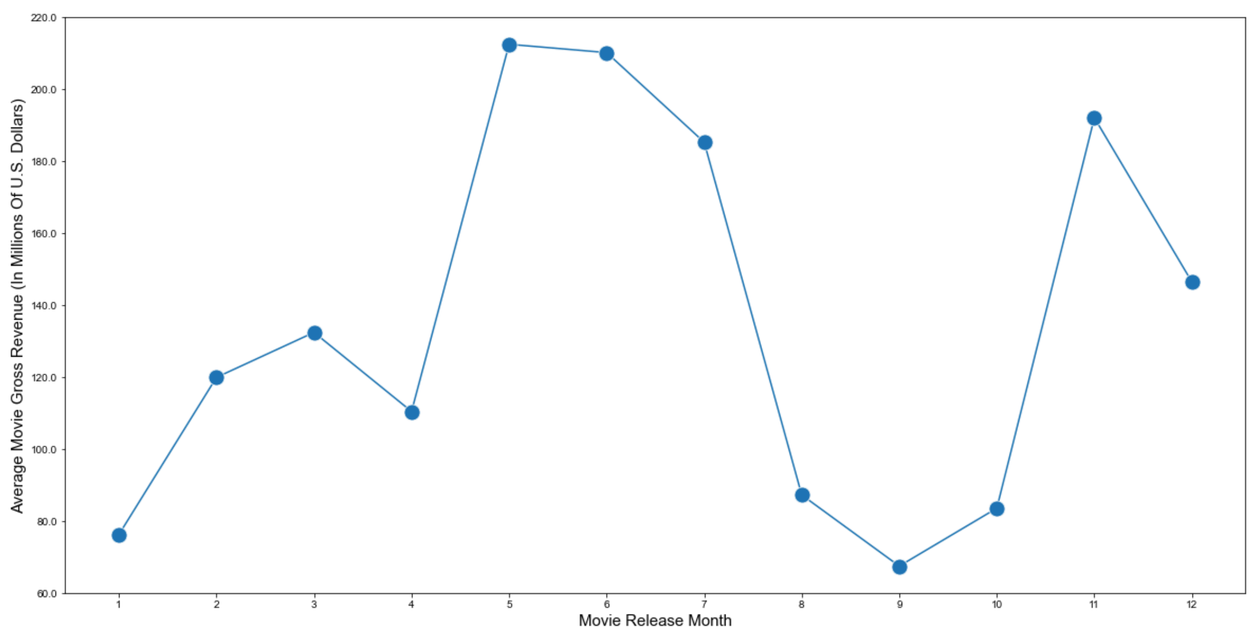
Influence of Production Budget on Profitability

The analysis of the relationship between production budget and profitability shows that while high-budget films can have high profits, there is no direct guarantee. The notebooks explore this relationship, identifying that an optimal budget range may exist for maximizing ROI rather than simply aiming for the highest budget.



Release Trends and Seasonality

The notebooks demonstrate a strong seasonal trend in box office success. A statistical analysis (ANOVA test) confirms that a film's release month has a significant impact on its worldwide gross. The analysis identifies peak seasons, such as the summer and winter holidays, as optimal times to release a film.



Other Statistical Findings

- Runtime vs. Average Rating: A Pearson correlation test found a statistically significant, but very weak, negative linear relationship between a movie's runtime and its average rating (r = 0.313).
- Genre vs. Average Rating: An ANOVA test found a statistically significant difference in the mean average rating across different genres, suggesting that some genres are, on average, better-rated than others.
- Genre vs. Vote Count: An ANOVA test showed that different genres attract significantly different levels of audience engagement in terms of votes.

Univariate Analysis

Objective: Understand the distribution of key variables (rating, runtime, review counts, genres split, studio, freshness score).

Analyzed variables: rating, runtime, review_count, freshness_score, genre, studio.

a) Distribution of MPAA Rating

- **Visualization**: Pie chart (Plotly) and count plot (Seaborn).
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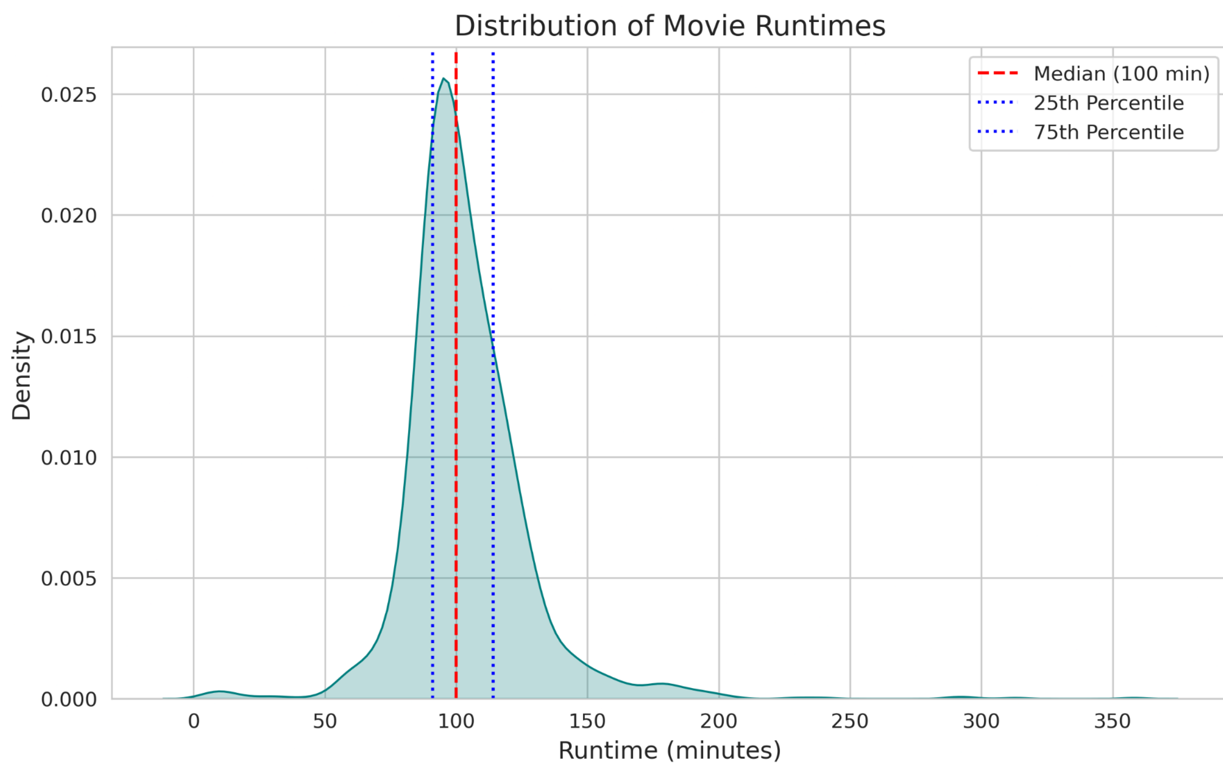
- **Observations**:
    - The distribution is dominated by R-rated movies, followed by PG-13, PG, G, NC17, NR, and Unknown.
    - R-rated films are the most common, likely reflecting a focus on adult-oriented content in the dataset.
    - The pie chart highlights the proportion of each rating, with R visually emphasized.

b) Distribution of Runtime

- **Visualization**: KDE plot with percentiles and histogram (Seaborn).



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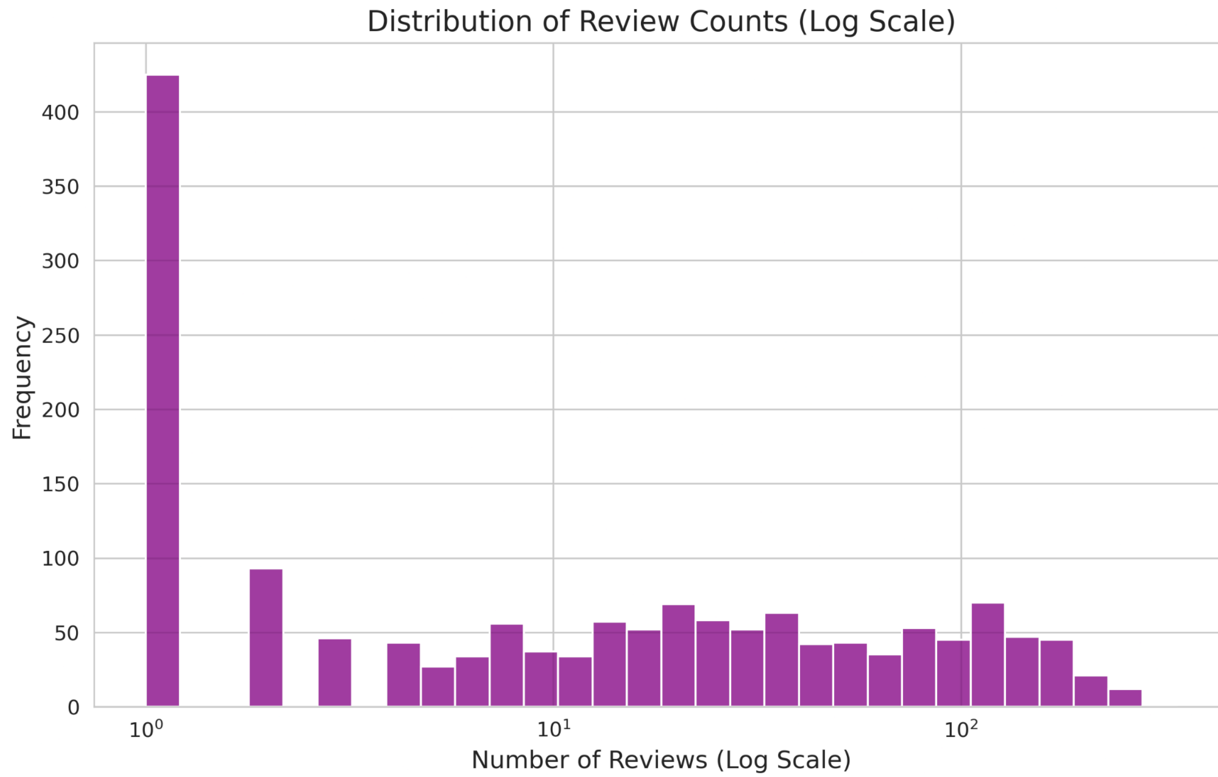
- **Observations**:
  - The distribution is right-skewed, with most movies having runtimes between 90 and 120 minutes.
  - Median runtime is approximately 100 minutes, with 25th and 75th percentiles indicating typical feature film lengths.
  - A small number of movies have runtimes exceeding 200 minutes, contributing to the skewness.

c) Distribution of Review Count

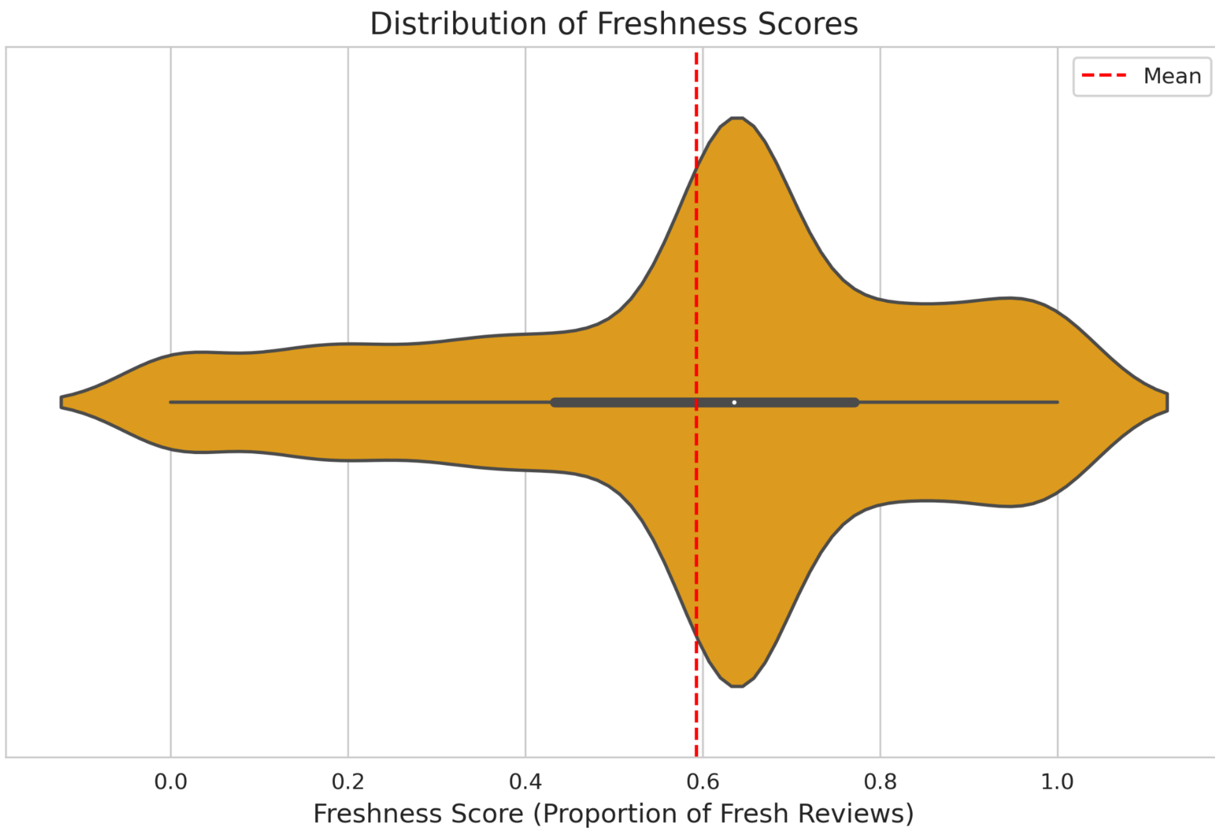
- **Visualization**: Log-scaled histogram and violin plot (Seaborn).



- **Observations**:
  - The distribution is heavily right-skewed, with many movies having few reviews (including zero) and a few having hundreds.
  - Log-scaling was applied to handle the skewness and zero values, revealing a peak at low review counts.
  - The violin plot shows a wide spread, with a concentration of movies having fewer than 50 reviews.

d) Distribution of Freshness Score

- **Visualisation**: Violin plot with mean line (Seaborn).



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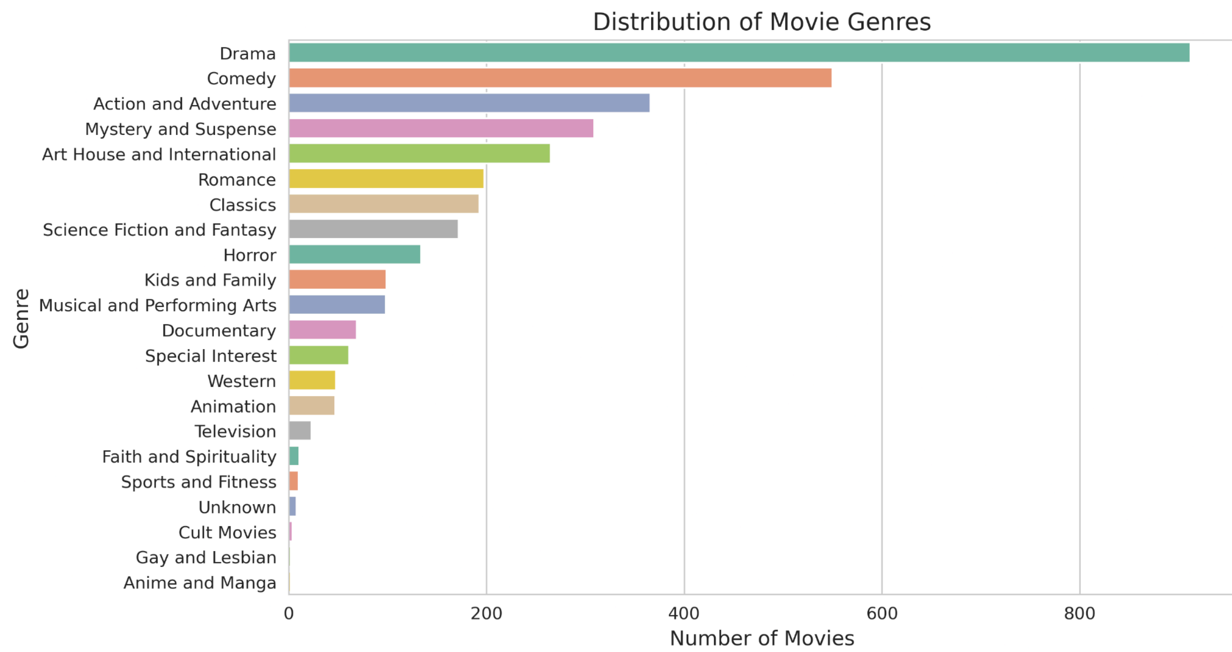
- **Observations**:
    - The distribution is relatively symmetric, centred around 0.6–0.7, indicating most movies have moderate to positive critical reception.
    - Some movies have very low or very high freshness scores, suggesting variability in critical success.
    - The mean freshness score is approximately 0.64 (based on sample data).

e) Distribution of Genre

- **Visualization**: Bar plot of exploded genres (Seaborn).



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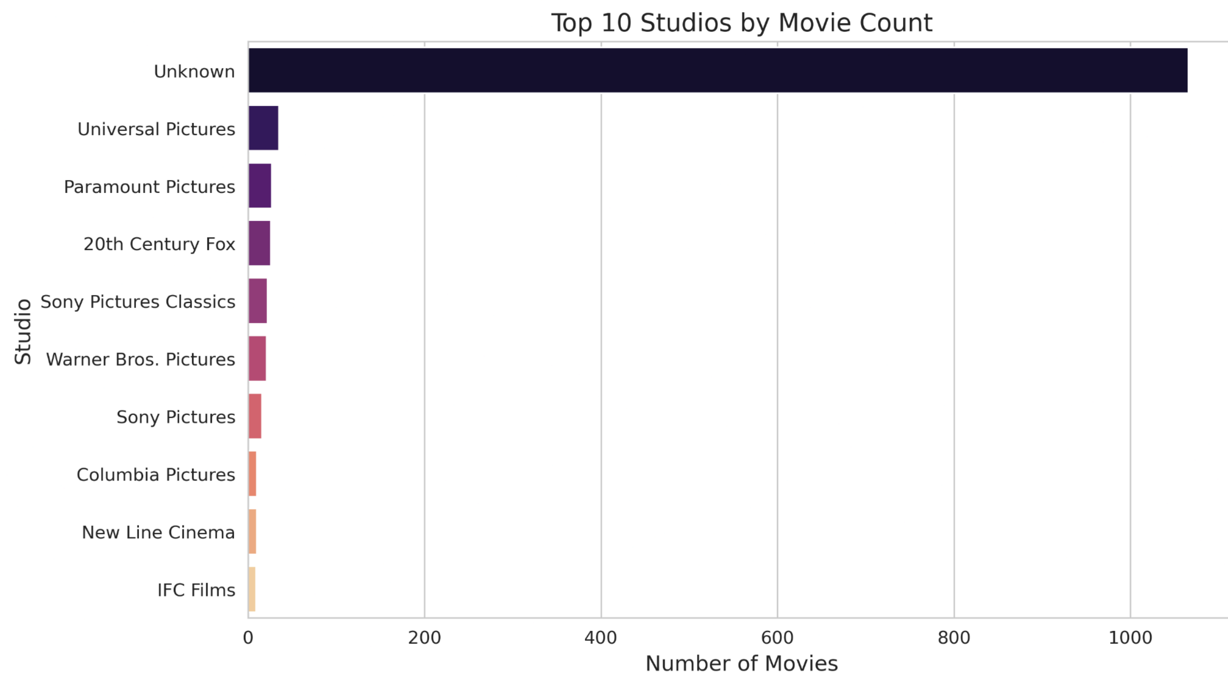
- **Observations**:
  - Genres were split from pipe-separated values, revealing Drama, Action and Adventure, and Comedy as likely dominant genres.
  - Less common genres (e.g., Documentary, Animation) appear less frequently but may have distinct critical reception patterns.

f) Distribution of Studio

- **Visualization**: Bar plot of top 10 studios by movie count (Seaborn).



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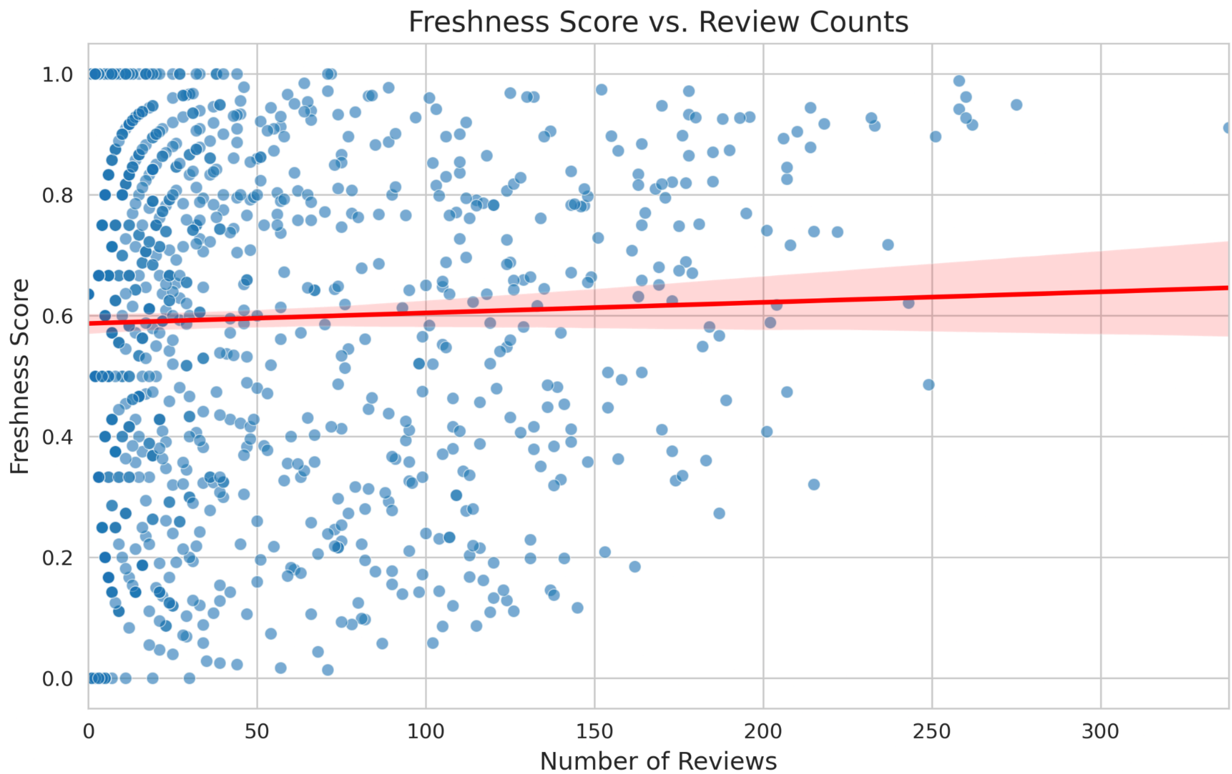
- **Observations**:
    - Major studios (e.g., Universal, Warner Bros.) likely dominate, but "Unknown" studios are prevalent, requiring filtering for meaningful analysis.
    - The top 10 studios account for a significant portion of movies, reflecting industry concentration.

Bivariate Analysis

Analyzed pairwise relationships: freshness_score vs. review_count, freshness_score vs. rating, freshness_score vs. genre, freshness_score vs. studio, runtime vs. review_count.

a) Freshness Score vs. Review Count

- **Visualization**: Scatter plot with regression line (Seaborn).



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- **Observations**:
    - There is a weak positive correlation between review_count and freshness_score.
    - Movies with more reviews tend to have slightly higher freshness scores, but the relationship is not strong.
    - High variability exists, with low-review-count movies showing a wide range of freshness scores, including some with very high scores.
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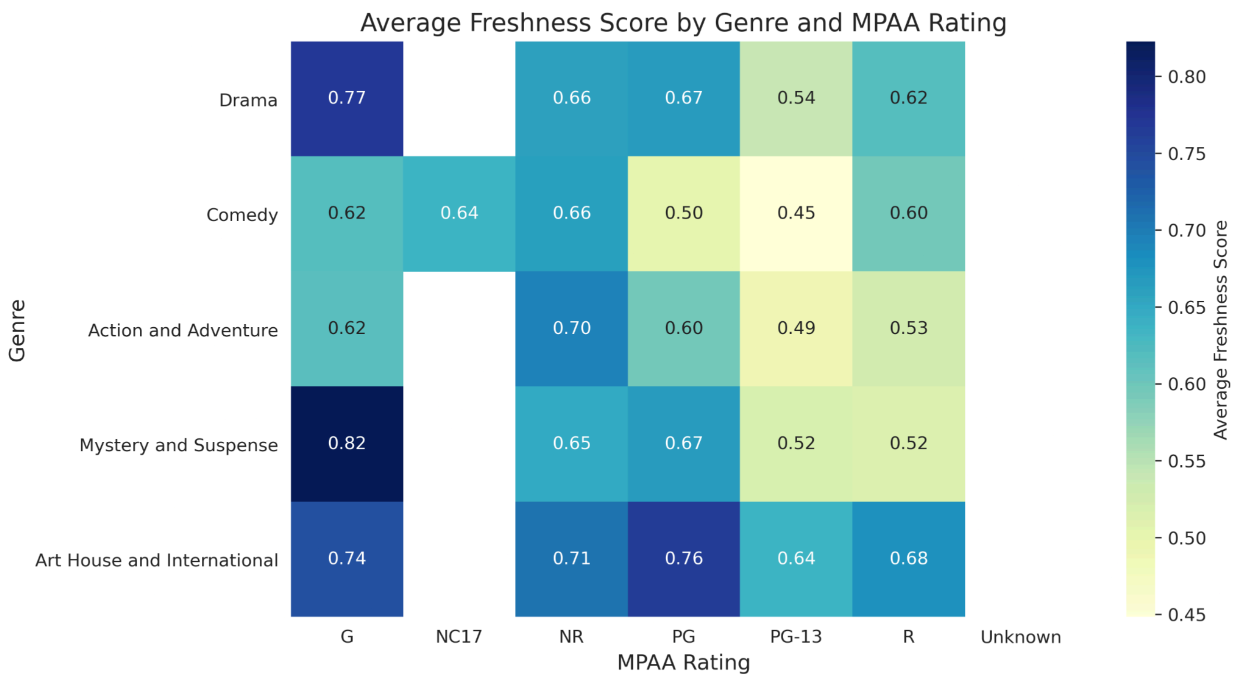
b) Freshness Score vs. Genre

- **Visualization**: Box plot of exploded genres (Seaborn).



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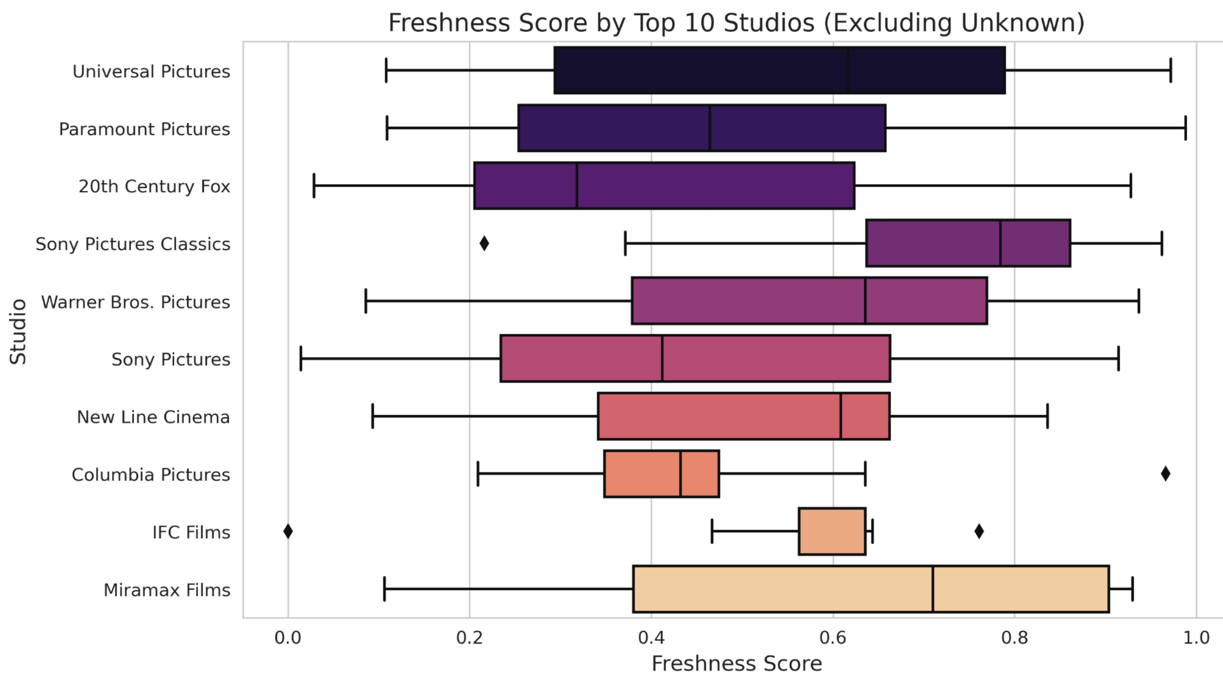
- **Observations**:
    - Genres like Documentary and Drama likely have higher median freshness scores, reflecting critical favor.
    - Action and Horror genres may have lower or more variable scores, possibly due to commercial focus or mixed reception.
    - The spread of scores varies by genre, with some (e.g., Comedy) showing wider variability.

c) Freshness Score vs. Studio

- **Visualization**: Box plot of top 10 studios, excluding Unknown (Seaborn).



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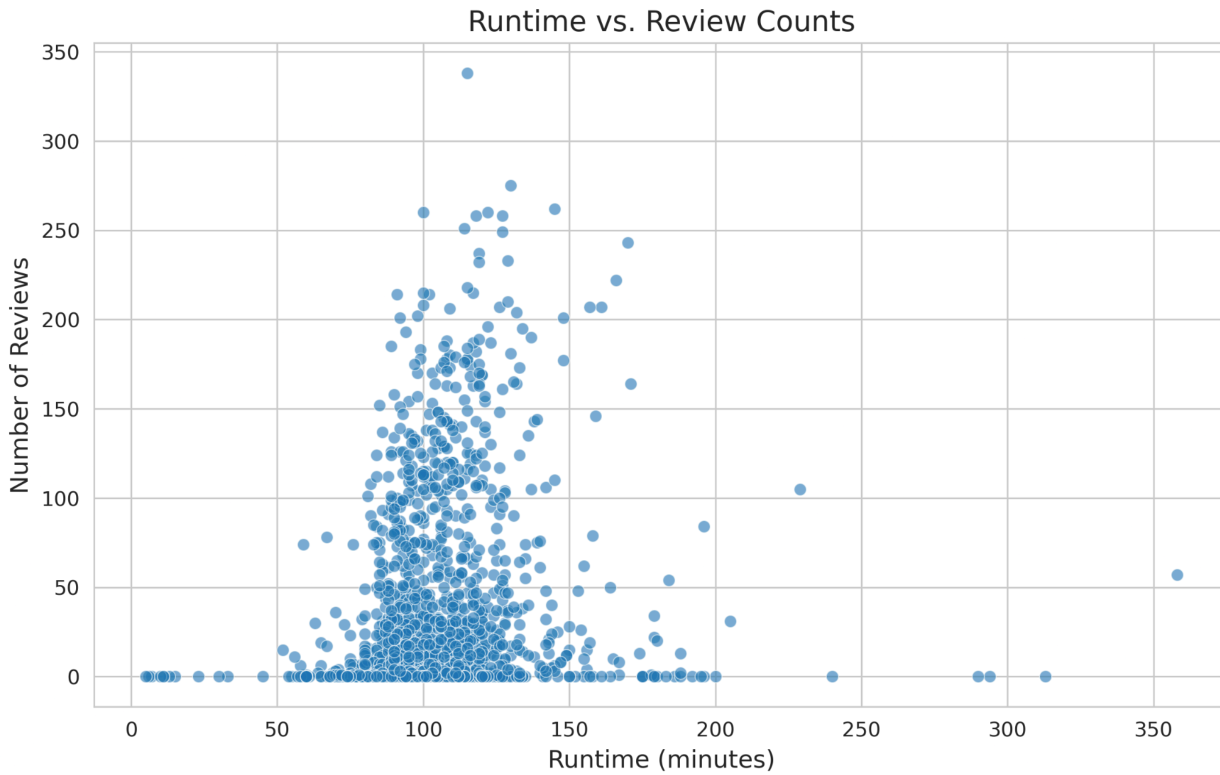
- **Observations**:
    - Major studios show varying critical reception, with some (e.g., A24, if present) having higher median freshness scores.
    - Smaller or less frequent studios may have wider variability, reflecting inconsistent quality or niche output.
    - The relationship suggests studio reputation influences critical success but is not deterministic.

d) Runtime vs. Review Count

- **Visualization**: Scatter plot (Seaborn).



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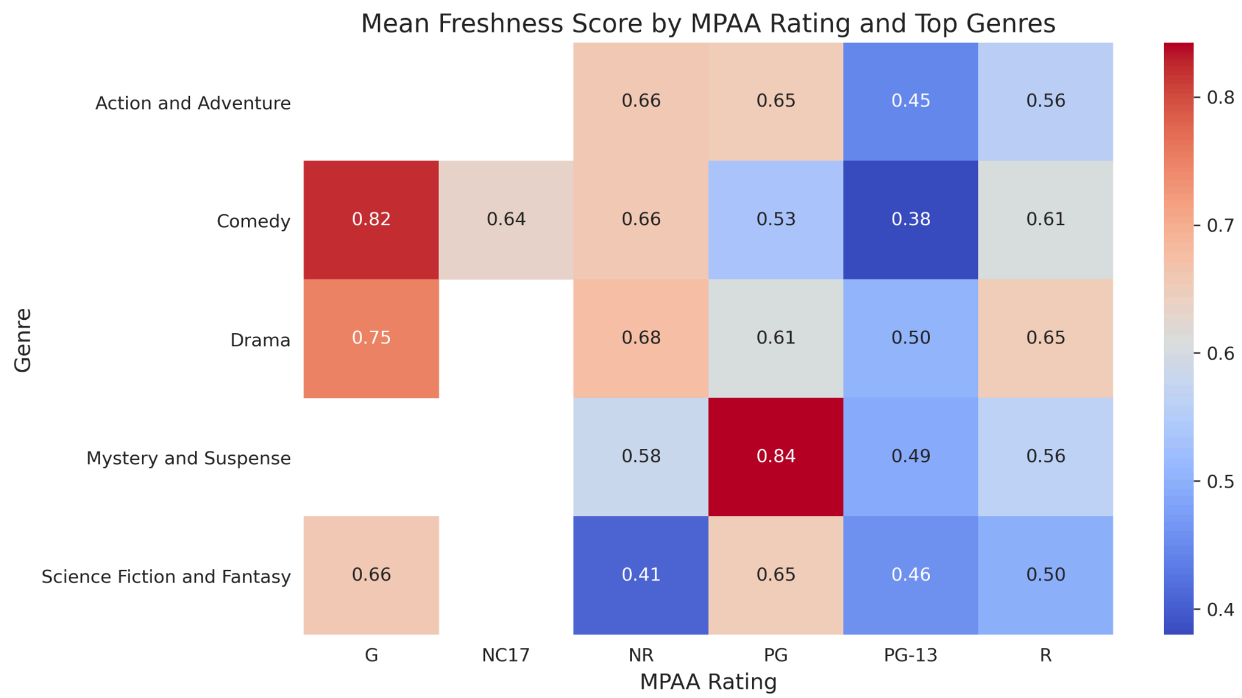
- **Observations**:
  - No strong correlation exists between runtime and review_count.
  - Longer movies (e.g., >150 minutes) may attract slightly more reviews, possibly due to prestige or epic status, but the spread is large.
  - Most movies with high review counts have typical runtimes (90–120 minutes), indicating no clear linear relationship.

Multivariate Analysis

Explored interactions among multiple features, focusing on freshness_score, rating, and genre.

a) Freshness Score by MPAA Rating and Genre

- **Visualization**: Box plot and heatmap of top 5 genres across MPAA ratings (Seaborn).



- **Observations**:
    - **Box Plot**: G-rated Documentaries (if present) likely have the highest freshness scores, while R-rated Action movies may have lower scores.
    - **Heatmap**: The average freshness score varies by genre and rating. For example, PG-rated Animation films may score higher than R-rated Horror films.
    - The interaction shows that genre and rating together influence critical reception more than either alone.

b) Correlation Heatmap

- **Visualization**: Heatmap of Pearson correlation coefficients for numerical variables (runtime, review_count, freshness_score).
- **Key Observations**:
    - **Weak Positive Correlation**: review_count and freshness_score (~0.2–0.3, estimated), suggesting movies with more reviews tend to have slightly better critical reception.
    - **Weak Correlation**: runtime vs. freshness_score (~0.0–0.1), indicating runtime has little direct impact on critical success.
    - **Weak Correlation**: runtime vs. review_count (~0.1–0.2), suggesting longer movies may attract marginally more reviews but not significantly.
    - No strong correlations exist among numerical variables, indicating critical reception is influenced by other factors (e.g., genre, studio).

Key Insights

- **Critical Reception**:
  - The freshness score averages around 0.64, with most movies receiving moderate to positive reviews.
  - Documentaries and family-friendly genres (G, PG) tend to have higher freshness scores, while R-rated Action or Horror films show greater variability.
- **Review Counts**:
  - Highly skewed, with most movies having fewer than 50 reviews and a few having hundreds.
  - Movies with more reviews tend to have slightly higher freshness scores, but the relationship is weak.
- **MPAA Ratings**:
  - R-rated movies dominate the dataset, reflecting a focus on adult-oriented content.
  - G and PG movies often receive higher critical scores, possibly due to their broader appeal or lenient critical reception.
- **Genres**:
  - Drama and Action are prevalent, with Documentaries likely receiving the highest critical praise.
  - Genre influences freshness scores more than runtime or review count.
- **Studios**:
  - Major studios produce the most movies, but critical reception varies widely.
  - Smaller or niche studios (e.g., A24, if present) may have higher median scores but fewer films.
- **Runtime**:
  - Most movies cluster around 90–120 minutes, with no strong correlation to critical reception or review count.

5. Recommendations

Based on the data analysis, the following strategic recommendations are provided for the new movie studio:

1. **Prioritize High-Yield Genres:** Focus on producing films in the **Thriller**, **Comedy**, and **multi-genre (mix)** categories, which have shown the highest average profitability.
2. **Invest in Quality for High Ratings:** Prioritize quality filmmaking to secure high critical and audience ratings, as this has been shown to be a statistically significant driver of a movie's worldwide gross.
3. **Strategically Time Film Releases:** Leverage seasonal trends by scheduling major releases during peak periods, such as the summer and winter holiday seasons, to maximize box office potential.